USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[22CSBS229]**

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Draw neat sketches wherever necessary

ANSWER ALL QUESTIONS

<table>
<tr><td></td><td></td><td colspan="2">PART-A</td><td colspan="2">05 X 02=10</td></tr>
<tr><td>1.</td><td>a.</td><td>Define Operations research (OR) with an application.</td><td>L1</td><td>CO1</td><td>02</td></tr>
<tr><td></td><td>b.</td><td>Verify whether the following transportation problem is balanced or not? If not, convert into a balanced transportation problem.
<table>
<tr><td>Source</td><td colspan="3">Destination</td><td>Availa bility</td></tr>
<tr><td></td><td>A</td><td>B</td><td>C</td><td></td></tr>
<tr><td>1</td><td>15</td><td>25</td><td>15</td><td>200</td></tr>
<tr><td>2</td><td>16</td><td>18</td><td>35</td><td>300</td></tr>
<tr><td>Requirement</td><td>120</td><td>150</td><td>130</td><td></td></tr>
</table></td><td>L2</td><td>CO3</td><td>02</td></tr>
<tr><td></td><td>c.</td><td>At a one-window drive-in bank, customers arrive following a Poisson distribution with a mean of 10 per hour. The service time per customer follows an exponential distribution with a mean of five minutes. Find the utilization factor of the drive-in bank?</td><td>L3</td><td>CO4</td><td>02</td></tr>
<tr><td></td><td>d.</td><td>Describe critical path in project network.</td><td>L2</td><td>CO5</td><td>02</td></tr>
<tr><td></td><td>e.</td><td>Describe the following:
i. Ordering cost
ii. Shortage cost</td><td>L2</td><td>CO6</td><td>02</td></tr>
<tr><td></td><td></td><td colspan="2">PART-B</td><td colspan="2">06 X 05=30</td></tr>
<tr><td>2</td><td>a.</td><td>Outline the phases involved in addressing an Operations Research problem</td><td>L2</td><td>CO1</td><td>05</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td></td><td>b.</td><td>A manufacturing company makes two models A and B of a product. Each piece of Model A requires 9 labour hours for fabricating and 1 labour hour for finishing. Each piece of Model B requires 12 labour hours for fabricating and 3 labour hours for finishing. For fabricating and finishing, the maximum labour hours available are 180 and 30 respectively. The company makes a profit of Rs 8000 on each piece of model A and Rs 12000 on each piece of Model B. Formulate this problem using Linear Programming Problem.</td><td>L2</td><td>CO2</td><td>05</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>3</td><td>a.</td><td>Outline the assumptions of linear programming problem (LPP).</td><td>L2</td><td>CO2</td><td>05</td></tr>
</table>

		OR			
	b.	Formulate the dual of the following primal problem: $MinZ = 15x_1 + 45x_2$ $Subjectto$ $2x_1 + 9x_2 = 20$ $15x_1 + 3x_2 = 15$ $13x_1 + 15x_2 = 45$ $x_1, x_2 \geq 0$	L2	CO2	**05**
4	**a.**	Formulate a linear programming model for the following transportation problem involving 3 sources and 3 destinations.	L2	CO3	**05**

	D_1	**D_2**	**D_3**	**Supply**
S_1	3	8	5	15
S_2	4	4	2	10
S_3	6	5	8	15
Demand	15	5	20	

		OR			
	b.	Determine the initial basic feasible solution for the following transportation problem using north-west corner method:	L3	CO3	**05**

	D_1	**D_2**	**D_3**	**D_4**	**Supply**
S_1	11	13	17	14	250
S_2	16	18	14	10	300
S_3	21	24	13	10	400
Demand	200	225	275	250	

5	**a.**	Explain the typical behaviour exhibited by customers while waiting in a queue.	L2	CO1	**05**
		OR			
	b.	Write R code to generate random numbers from a uniform distribution between 5 and 15.	L3	CO4	**05**
6	**a.**	Discuss why project management is important and list few applications.	L2	CO5	**05**
		OR			
	b.	Construct a network for the project whose activities and their precedence relationships are given below.	L3	CO5	**05**

Activity	A	B	C	D	E	F	G	H	I
Predecessor	-	-	A, B	B	B	A, B	F, D	F, D	C, G

7	**a.**	The demand of an item for an industry is 9000 units/year. The ordering cost is Rs. 15 per order and the carrying cost per unit per year is Rs. 3. Determine the economic order quantity (EOQ).	L2	CO6	**05**
		OR			
	b.	Discuss the inventory theory contributing to operational efficiency in organizations. Also, list its applications.	L2	CO6	**05**
		PART-C		**06 X 10=60**	
8	**a.**	Solve the following LPP using graphical method: $MaximizeZ = 2x_1 + 4x_2$ $Subjectto$ $x_1 + 2x_2 \leq 5$ $x_1 + x_2 \leq 4$ $x_1, x_2 \geq 0$ Explain the graphical solutios.	L3	CO2	**10**
		OR			
	b.	Explain the following terminologies: **i.** Feasible solution **ii.** Feasible region **iii.** Optimal solution **iv.** Infeasible solution	L2	CO2	**10**
9	**a.**	Solve the following LPP using Simplex method: $MaxZ = 6x + 8y$ $Subjectto$ $5x + 10y \leq 60$ $4x + 4y \leq 40$ $x, y \geq 0$	L3	CO2	**10**
		OR			
	b.	Write R program to solve the following LPP: $MinimizeZ = 20x + 10y$ $Subjectto$ $x + 2y \leq 40$ $3x + y \geq 30$ $4x + 3y \geq 60$	L3	CO2	**10**

$x, y \geq 0$

10 a Write R program to solve the following Transportation problem: — L3 — CO3 — **10**

		Destinations			
		1	2	3	**Supply**
Sources	1	20	20	30	100
	2	40	10	20	150
	3	10	30	10	400
	Demand	200	150	300	650

OR

B Use the Hungarian method to solve the following assignment problem: — L3 — CO3 — **10**

	Machines				
Jobs	*A*	*B*	*C*	*D*	*E*
1	13	8	16	18	19
2	9	15	24	9	12
3	12	9	4	4	4
4	6	12	10	8	13
5	15	17	18	12	20

11 a. In a railway marshalling yard, goods trains arrive at a rate of 30 trains per day. Assuming that inter-arrival time and service time distribution follows an exponential distribution with an average of 30 minutes, calculate the following. — L3 — CO4 — **10**

- **i.** The mean queue size.
- **ii.** The probability that queue size exceeds 10.
- **iii.** If the input of the train increases to an average of 33 per day, what will be the changes in (i) and (ii)?

OR

b. Discuss the characteristics of Queuing system in detail. — L2 — CO4 — **10**

12 a. A project schedule has the following characteristics: — L3 — CO5 — **10**

Activity	1-2	1-3	1-4	2-3	3-4	2-6	3-5	5-6	6-8	5-8	4-7	5-7	7-8
Time(Days)	4	5	3	3	4	2	6	5	7	6	4	4	8

From the above information:

- **i.** Construct a network diagram.
- **ii.** Compute the earliest event time and latest event time.
- **iii.** Determine the critical path and total project duration.
- **iv.** Compute total float for each activity

		OR			
	b.	The following table shows the jobs of a network along with their time estimation **Job** | 1-2 | 1-6 | 2-3 | 2-4 | 3-5 | 4-5 | 6-7 | 5-8 | 7-8 **a (Optimistic)** | 1 | 2 | 2 | 2 | 7 | 5 | 5 | 3 | 8 **m (Most likely)** | 7 | 5 | 14 | 5 | 10 | 5 | 8 | 3 | 17 **b (Pessimistic)** | 13 | 14 | 26 | 8 | 19 | 17 | 29 | 9 | 32 **i.** Construct the project network **ii.** Determine the expected project completion time **iii.** Find the probability of the project completing in 40 days.	L3	CO5	**10**
13	**a.**	The annual demand of the raw material for a company is 9000 units. The ordering cost is Rs. 100/order and the carrying cost is Rs. 2.4/unit/year, where the purchase price is Rs. 1/unit. Determine the following: **i.** Economic order quantity (EOQ) **ii.** Total cost w.r.t EOQ **iii.** Number of orders per year **iv.** Time between consecutive two orders	L3	CO6	**10**
		OR			
	b.	The demand of a component for an industry is 12,000 units/year. The carrying cost is Rs. 2/unit/year, the ordering cost is Rs. 600/order and the shortage cost is Rs. 10/unit/year. Determine the following: **i.** Economic order quantity (EOQ) **ii.** Number of orders per year **iii.** Maximum inventory **iv.** Maximum shortage quantity	L3	CO6	**10**